Monero: A Comprehensive Guide to What It Is and How It Works

12-15 minutes ⋮ 12/7/2023

The Monero network allows users to execute transactions without revealing their identities, transaction history, or the exact sums involved. The network and native crypto, XRM, offer a secure haven for those looking for privacy in an era marked by increasing scrutiny and documentation of financial transactions. **This guide will walk you through everything you need to know about the network, its operational principles, the native Monero crypto, and its role within the wider crypto landscape.**

In this guide:

- What is Monero?
- History of Monero crypto
- How does Monero work?
- What makes Monero unique?
- How Monero differs from Bitcoin
- Monero crypto (XMR)
- Tokenomics
- XMR wallets
- Monero's path forward
- Frequently asked questions

What is Monero?

Monero is an open-source cryptocurrency that prioritizes privacy and champions censorship resistance. Unlike better-known cryptocurrencies like Bitcoin and Ethereum, Monero utilizes advanced technologies to safeguard user privacy, ensuring that transactions remain confidential and

untraceable. This unique feature makes Monero the sole major cryptocurrency where every user enjoys automatic, by-default anonymity**.**

All transactions on Monero are confidential and untraceable. **This anonymity is achieved through three key technologies: Stealth Addresses, Ring Signatures, and RingCT, which hide the sender, receiver, and transaction amount.**

History of Monero crypto

Monero's history is closely tied to that of Bitcoin. **The network originated in 2012 as a fork of Cryptonote, a project created to improve privacy in the cryptosphere** and, in particular, address what the founders saw as Bitcoin's limitations.

The idea was that Crytponote could make cryptocurrency transactions unlinkable and untraceable, **akin to traditional cash transactions.**

> Classic Bitcoin addresses, once being published, become unambiguous identifier for incoming
> payments, linking them together and tying to the recipient's pseudonyms. If someone wants to
> receive an "untied" transaction, he should convey his address to the sender by a private channel.
> If he wants to receive different transactions which cannot be proven to belong to the same owner
> he should generate all the different addresses and never publish them in his own pseudonym.
>
> **We propose a solution allowing a user to publish a single address and receive unconditional**
> **unlinkable payments.**
>
> *Nicolas van Saberhagen: Cryptonote v2.0 Whitepaper*

Originally named Bit Monero ("monero" implies coin in Esperanto), the network was forked and renamed Monero after it lost community trust. This erosion of confidence stemmed from some questionable actions by the

network's development team, including the stealthy pre-mining of a significant portion of the coin supply.

Monero's group of developers, much like Bitcoin's enigmatic Satoshi Nakamoto, remain largely anonymous, using pseudonyms. Notably, the developers did not reserve ownership for themselves upon introducing XMR, relying instead on community contributions and support for its ongoing development.

While the majority of the team's identities remain unknown, a confirmed member of the founding team is Riccardo Spagni (aka @FluffyPony) — who, in March 2023, was forced to deny claims posted on X that he had been working as an informant for Interpol.

How does Monero work?

Monero operates on a proof-of-work consensus mechanism, meaning it depends on miners solving complex computational algorithms to confirm transactions and introduce new blocks on the mainnet.

The platform's is built on open-source CryptoNote code, which also forms the basis of some lesser-known cryptos. Let's look at the main technologies that keep Monero transactions private in a little more detail.

1. **Stealth addresses:** Monero incorporates the concept of stealth addresses to boost privacy. These addresses enable unique, one-time public addresses for each recipient, ensuring each transaction has a unique public address that cannot be linked to the recipient. Despite this, the recipient can publish only one inward address while sending all their transactions to unique addresses on the blockchain.

2. **Ring signatures:** Monero employs ring signatures to obscure the identities of those involved in transactions. When a Monero transaction takes place, the outside observer can only see a pool of potential senders. This makes it impossible to pinpoint the actual sender. This powerful feature makes Monero transactions truly private.

These privacy-enhancing technologies work together to keep Monero transactions shielded from public scrutiny. Notably, they do not just offer anonymity but also improve fungibility.

What makes Monero unique?

Monero introduced several key features and upgrades to ensure privacy, anonymity, and functionality. As well as stealth addresses and ring signatures, these include:

- **Tail Emission:** A perpetual emission of coins to ensure sustainability without excessive inflation.
- **Ring CT (Ring Confidential Transactions):** This technology conceals transaction amounts.
- **Bulletproofs:** Introduced in 2018 or 2019, bulletproofs significantly reduced transaction sizes and fees.
- **RandomX:** An egalitarian mining algorithm that promotes fair distribution by emphasizing “one CPU, one vote.”
- Easy mining **process:** Monero’s key advantage lies in its strong commitment to decentralization, particularly in its mining process. Unlike Bitcoin, which relies on expensive specialized machines (ASICs) for mining, Monero primarily uses regular CPUs found in everyday computers. **This approach makes mining more accessible to ordinary individuals and helps prevent centralization and potential network vulnerabilities.**

How Monero differs from Bitcoin

To grasp the significance of Monero, it’s important to understand how it differs from Bitcoin, the pioneering cryptocurrency.

1. **Absolute privacy:** Monero distinguishes itself from Bitcoin by offering complete privacy and anonymity in transactions. Unlike Bitcoin, which has some traceability and transaction history tagging, Monero ensures that all transactions remain confidential and untraceable.

2. **Block size:** Monero employs a flexible approach to block sizes. This is unlike Bitcoin’s fixed limit, which could eventually lead to increased transaction fees. Monero allows block sizes to increase as needed to avoid network congestion. Monero imposes penalties for excessively large blocks to prevent abuse, which helps ensure smoother network operation.

3. **Decentralization:** Monero promotes a higher level of decentralization through its mining mechanism. Unlike Bitcoin, which relies on specialized ASICs, Monero mining primarily uses common CPUs, making it accessible to average users. This decentralization reduces the risk of centralization and enhances network robustness and resilience against potential adversarial interests.

4. **Fungibility:** In contrast to Bitcoin, Monero coins are indistinguishable and untraceable, ensuring full fungibility. This appeals to users who value the privacy and fungibility of their currency, as every Monero coin has the same value and history.

Monero crypto (XMR)

$XMR is used as a native currency within the Monero network. As of Dec. 7, 2023, $3.2B, the Monero crypto is ranked 26th in the global crypto ratings. Furthermore, it has an average daily trading volume of **$78.7M.**

Monero crypto (XMR) price chart: BeInCrypto

XMR can be used for peer-to-peer payments and value storage within the Monero network. The native coin is also used to pay fees for transactions. **Due to its robust privacy features, offering anonymity by default, $XMR has found significant usage as a black or grey market currency.**

Our expert Monero price prediction puts the potential future price of XMR at a maximum of $237.58 in 2024, $261.34 in 2025, and $287.47 in 2026. Note that predictions can never be completely accurate and should only be used as a guideline. While fundamental and technical analysis can provide solid groundwork, news cycles, wider market forces, and social sentiment can all impact the price of a crypto, XMR included.

Users looking to buy monero can purchase the XMR coin on a number of leading exchanges, including OKX (best for spot and copy trading), Kraken (best for security features), and Binance (most comprehensive overall ecosystem).

Tokenomics

Similar to Bitcoin and Litecoin, Monero block rewards decrease over time. Monero (XMR) operates via a set of tokenomics characterized by a focus on privacy and limited inflation. As of December 2023, Monero has a block reward of 0.6 XMR, translating to approximately 0.3 XMR generated every minute. **This relatively low inflation rate results in roughly $80 of new XMR entering circulation each minute.**

Monero's emission rate was carefully designed to ensure fair coin distribution as the network evolved. In its early days, the emission rate was relatively high and gradually decreased as the network matured**. This approach encouraged early adopters, such as miners, while also preventing a sudden flood of XMR coins into the market, which could have threatened its intrinsic value.**

Governance

Monero's governance happens off-chain, with improvement proposals submitted on GitHub and subject to community voting. Upon consensus and approval, the code is integrated into the Monero Core Client, though node updates can be slow due to the decentralized nature. To address this, Monero schedules **biannual hard** forks, encouraging node operators to stay engaged and ensuring timely updates. The most recent hard fork occurred on July 16, 2022.

XMR wallets

A Monero wallet is a digital repository designed to safeguard your Monero (XMR). Unlike a physical wallet, which holds tangible cash, a Monero wallet doesn't actually store your XMR coins. Instead, it houses the essential private keys that grant you access to your holdings and enable you to conduct transactions.

To manage and access your Monero assets, you have various Monero wallet options at your disposal. These include hardware wallets like Ledger and Trezor — cold wallets that provide offline security.

Alternatively, software wallets, also known as "hot wallets," come in both desktop and mobile versions. Solid options for storing XMR include Trust Wallet, Cake Wallet, and Feather Wallet.

Notably, the Monero GUI Wallet stands out as an open-source graphical user interface (GUI) wallet developed by the Monero community.

Monero's path forward

Monero, stands out as perhaps the most important and certainly most well-known privacy coin. The network provides a crucial service by empowering individuals to protect their financial privacy and retain control over their assets.

However, Monero's future is not guaranteed. Some consider it too obscure for institutional investors. Meanwhile, its robust privacy features are a concern to governments; the currency has the potential to facilitate the bypassing of restrictions and sanctions. This resistance to government control benefits Monero users but is likely to hinder broader adoption and possibly leave the network and XMR currency vulnerable to regulatory woes.

Frequently asked questions

Is Monero crypto a good investment?

Whether Monero (XMR) is a good investment depends on individual preferences and goals. XMR and the monero network offers a solid use case, providing privacy to its users. However, the crypto's value is subject to market volatility and can be impacted by regulatory changes, the news cycle, and wider market forces. Thorough research and risk assessment are essential before investing.

Is Monero better than Ethereum?

Monero and Ethereum cater to different needs. Monero is primarily designed for privacy-focused transactions, offering strong anonymity and fungibility. Ethereum, on the other hand, is versatile for smart contracts and DApps.

Is Monero risky?

Monero has inherent risks, including market volatility and regulatory uncertainty. A risk assessment is important for potential investors of any crypto. The actual network is built on solid blockchain technology and is secure.

Can Monero transactions be traced?

Monero transactions are designed to be private and untraceable. The network uses stealth addresses and ring signatures to achieve this. It is challenging to trace Monero transactions, as even with access to the blockchain, external observers cannot easily identify the sender, receiver, or transaction amount.

Disclaimer

In line with the Trust Project guidelines, the educational content on this website is offered in good faith and for general information purposes only.

BeInCrypto prioritizes providing high-quality information, taking the time to research and create informative content for readers. While partners may reward the company with commissions for placements in articles, these commissions do not influence the unbiased, honest, and helpful content creation process. Any action taken by the reader based on this information is strictly at their own risk.